

MAE 1001:

From numbers to loops: Unlocking Python's power



Introduction to Mechanical and Aerospace Engineering

Course website: <https://gwu-mae1001.github.io/fall2025/>

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Photo: Kartik Bulusu



“Study hard what interests you the most in the most undisciplined, irreverent and original manner possible.”

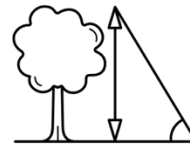
– Richard Feynmann



Computers calculate stuff



π



Icon Sources:

Floppy Disk by Diardha Gumi: <https://thenounproject.com/browse/icons/term/floppy-disk/>

harddisk by Start Up Graphic Design: <https://thenounproject.com/browse/icons/term/harddisk/>

calculations by Xinh Studio: <https://thenounproject.com/browse/icons/term/calculations/>

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Computers need some language to communicate

English ?

Primitive constructs

- Words
 - Alphabets
- Sentences with words

Python ?

Primitive constructs

Numbers, strings
Basic math operators

+
-
*
/
>=<



Python programs

- Sequences of definitions and commands
- Everything in Python is an **object**
 - **number 9**, is an object
 - **range()** is an object
- Manipulate **objects**
- **Objects:**
 - have a **type()**, that determine what python program can do with them

Analogy



type

human

- Python has two kinds of objects
 - **scalar**: very basic and other objects can be made with them
 - **non-scalar**: have an internal structure

Sources:

What is computation? By Ana Bell: <https://youtu.be/nykOeWgQcHM>

Bilbo Baggins: https://en.wikipedia.org/wiki/Bilbo_Baggins

Bilbo Baggins: https://heroes-and-villain.fandom.com/wiki/Bilbo_Baggins



Our focus is on scalar objects

int	Represent integers including 0 (whole numbers)
float	Represent real numbers (with decimals)
bool	Represent Boolean values True and False
NoneType	Special and has one value, None

Integer & Float operators

```
>>> type(9)
<class 'int'>
```

```
>>> type(9.0)
<class 'float'>
```

```
>>> type(range(9))
<class 'range'>
```

i+j	Sum of integers	} → i and j are integers. Result is an integer.
i-j	Difference	
i*j	Product	
i/j	Division	→ i and j are integers. Results in decimal. (float)
i%j	Remainder of i/j	} → i and j are integers. Result is an integer.
i**j	i raised to the power j	
i//j	Integer division	



Functions: A first look

A function is group of related statements that performs a specific task.

functions

- Break your program into smaller and modular chunks.
- Make a program more manageable and modular as a program grows larger and larger
- Avoid repetition and makes the code reusable

print()

- Example of a function you just learned
- Built-in function in Python

Functions are blocks of reusable pieces of code

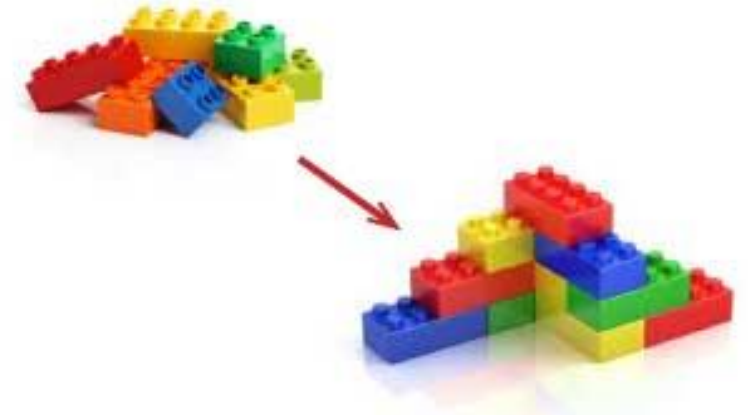


image source:
<https://www.thecoopformommies.com/teaching-colors-to-preschoolers/lego-blocks-clip-art-uidyo-clipart/>
<https://www.123rf.com/clipart-vector/lego.html?fromviewfrom=123rf/74381a74>



print()

More ways to use the print() function

sep

```
print("Hello", "World!", "I", "love", "Python", sep=",")
```

end

```
print("Hello", "World!", "I love Python", end=' ')
```

- By default, Python's print() function ends with a newline
- Commas between each entry outputs a space between each entry
- The entry of arguments of print() can be strings

0.1.2 - Strings

A *string* in Python is a sequence of letters, digits, or symbols (such as & or @) surrounded by either:

- A pair of double quotes, as in "Hello world!"
- A pair of single quotes, as in 'Hello world!'

Demos

An escape character is a backslash \ followed by the character you want to insert.

\'	Single Quote
\\	Backslash
\n	New Line
\r	Carriage Return
\t	Tab
\b	Backspace
\f	Form Feed
\ooo	Octal value
\xhh	Hex value



Variables

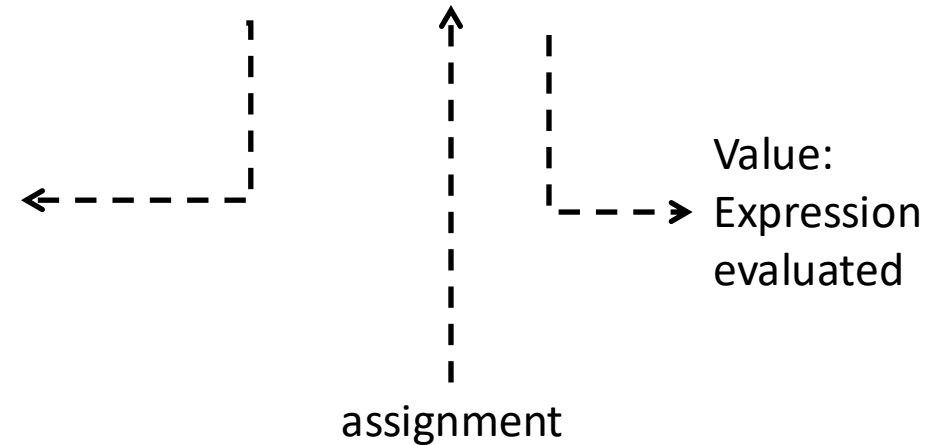
Give a name to the values of the expressions

The ***equal sign*** is an assignment of a value to a variable name.

- **Assignment binds the variable name to the value**
- Value is stored in the computer memory
- Invoke the variable name to retrieve the value
- **Variable refers to the value in the memory**

```
>>> pi_approx = 22/7
```

variable



Built-in function range()

Python's **range()** function
returns a sequence of
numbers
works only with integers

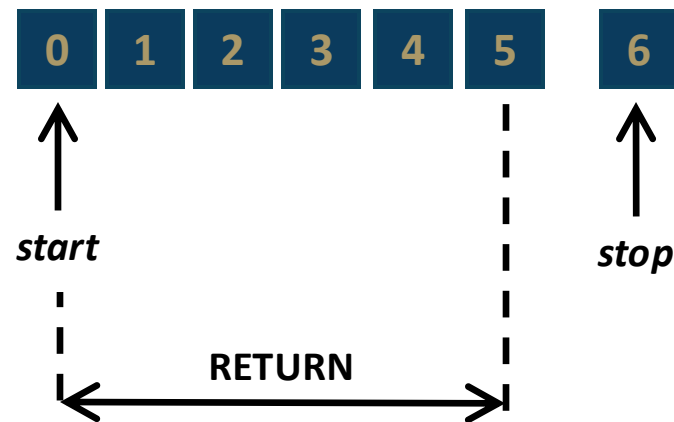
start:	at the value (default = 0)
step:	up or down at the increment value (default = 1)
stop:	at the value but not including it

range(stop)

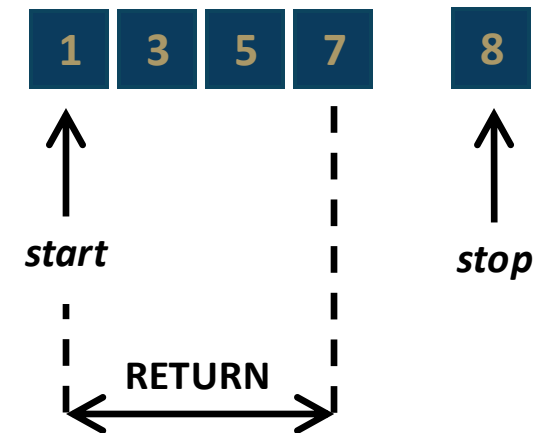
range(start, stop)

range(start, stop, step)

```
>>> range(6)
```



```
>>> range(1,8,2)
```



range()

- can be utilized in (for) loops
- to specify a range to iterate or do repetitions

Demos



What is a "loop" in computer programming ?

Loops_

codecademy



Repeating routines

From function to repeating functions

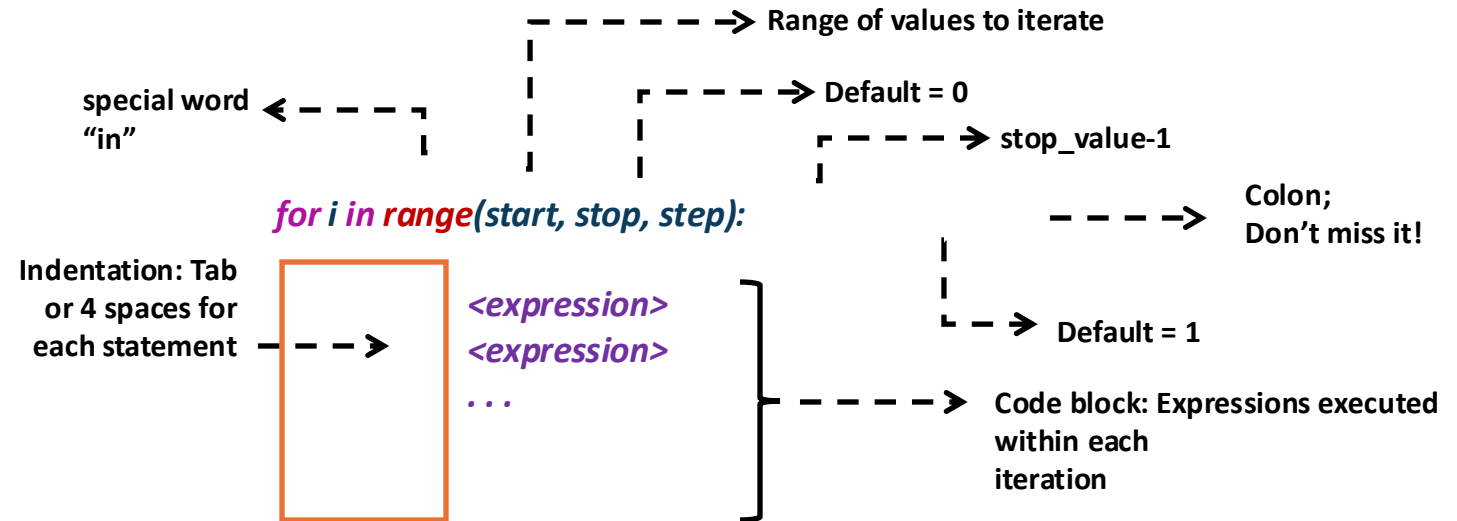
Stopping condition



Skeleton of the for-loop

```
for i in range(start, stop, step):
```

```
    <expression>  
    <expression>  
    ...
```

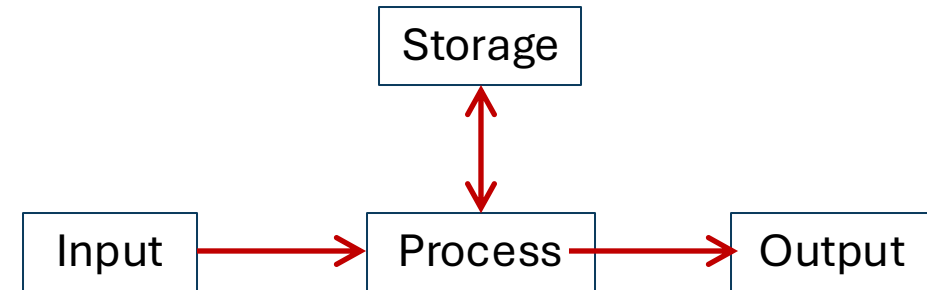


The Recipe Analogy

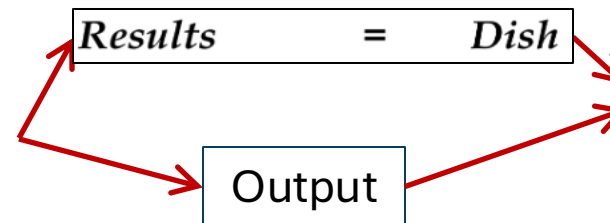
Program = *Recipe*

Laptop = *Cook*

Results = *Dish*
(the program's output)



**Counter-intuitively,
we start looking at**



print()

```
print("Hello World!")
print('Hello World!')
print("""George\nWashington was an
amazing person. A university was
named after him.""")
```

